

10 From diagram 1, torque = $F\left(\frac{l}{4} + \frac{l}{4}\right) = \frac{Fl}{2} = M$

In diagram 2, the perpendicular distance between the two forces F is $\frac{l}{2}$, which is the same in diagram 1. Hence the torque by the couple remains the same at M .

Answer: C

11 $P = Fv = (400)(20) = 8000 \text{ W}$